

Section 1. The short monitoring plan

This eleven question plan provides a solid foundation should you wish to develop further (See **Section 2. The extended monitoring plan**).

1 Why are you monitoring/surveying?

The first step in planning is to ask why do we want to monitor? Answers may vary but often groups simply want to know what the stream is like.

2 Who will use your information?

Potential users might include students and/or group members, etc. Name the main groups.

3 How will the information be used?

Data could be used for more than one purpose, e.g. to educate students about the principles of ecology or identify major trouble spots in the waterway. Knowing its main use will help determine the right kind of data to collect. Describe its intended use.

4 What will you monitor?

The things you choose to monitor will depend upon the question(s) you are asking as well as the resources available. For example, if your group wants to learn about the general ecological health of the waterway, then the main types of water bugs (macro-invertebrates) present could tell an interesting story. List the things that you will monitor. See **Table 1. General guidelines for monitoring** for a full list of things that can be monitored.

5 How accurate should the data be?

This depends upon the question(s) you are asking and how you intend to use the data. At the very least your data should be accurate enough to indicate the location of grossly polluted sites. Depending on your findings you may then choose to refine your monitoring program (see **Section 2. The extended monitoring plan**). For groups with a focus on education and raising awareness, the quality of the data is secondary to the actual process of collecting it.

6 What methods should be used?

This depends on your objective(s) and resources. There are often several ways of testing the same parameter, e.g. for high precision turbidity readings from 0 to 1000 NTU, a turbidity meter costing several thousand dollars is required, but a turbidity tube (≈ \$35) is suitable for less precise readings between 10 - 400 NTU. Use the same methods at all sites to allow comparison of data. List which methods will be used.

7 Where will you monitor?

The location of monitoring sites depends on whether you are monitoring a river, lake, estuary or ground-water, and also on the purpose of monitoring. For example, monitoring at a variety of typical sites in the catchment is good for providing information about its overall condition. On the other hand, sites located above and below a pollution source are needed to indicate its impact. Your sites should be representative of the condition of the waterway. Use a map to show your sites.



8 When will you monitor?

This depends on your resources and the purpose of monitoring. For example, if you are interested in a snapshot of the waterway, then monitor a number of sites on the same day. Monitoring pollution events, e.g. discharges from pipes, depends on the timing of the discharge and surveying the physical form of the stream is best done during low flows for safety reasons. Describe when you will monitor.

9 Who is going to be involved and how?

Indicate who will carry out surveys and/or test water samples, who will arrange transport to sites and back, who will prepare the water testing equipment to be used, who will photograph sites, etc.

10 How will monitoring data be managed and presented?

It is important to record and present the data. It helps to raise awareness of the condition of the waterway amongst members and helps you to refine your monitoring activities. Identify who will look after the data and describe how it will be managed.

11 How will you ensure that your information is credible?

Developing answers to the first ten questions above is the first step in conducting an effective visit to the waterway. For all surveys and tests, ensure that group members are adequately trained. For water quality tests, ensure that any instruments used are calibrated and read correctly and that any water samples from rivers are taken from the main current at about 25cm below the surface. List what you will do to improve the credibility of your information.

As you worked through the questions above, you may have found that a more thorough plan is needed to effectively gather data to answer the concerns of your group. In this case, go to **Section 2. The extended monitoring plan.**

